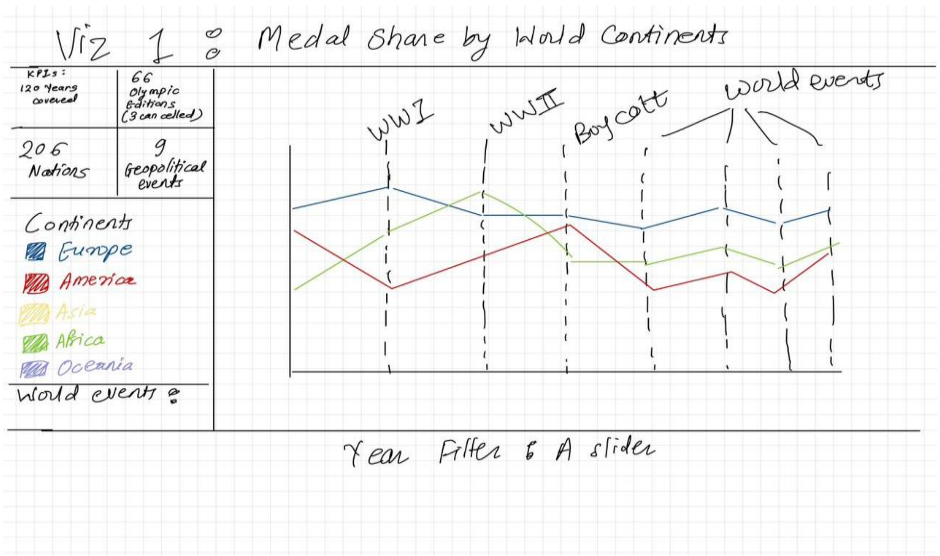
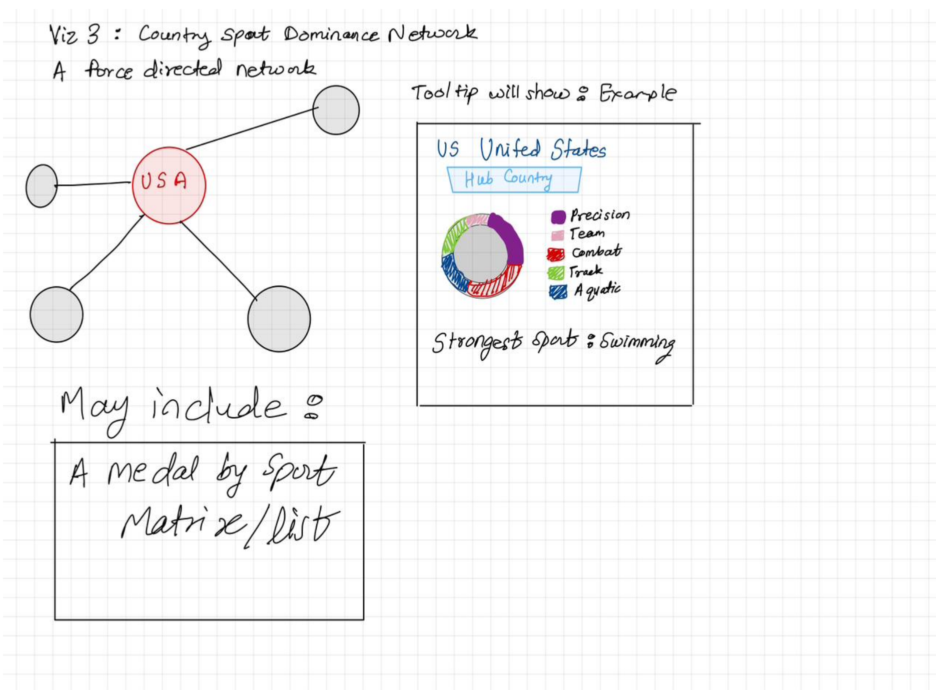
Namita Birari

Layout



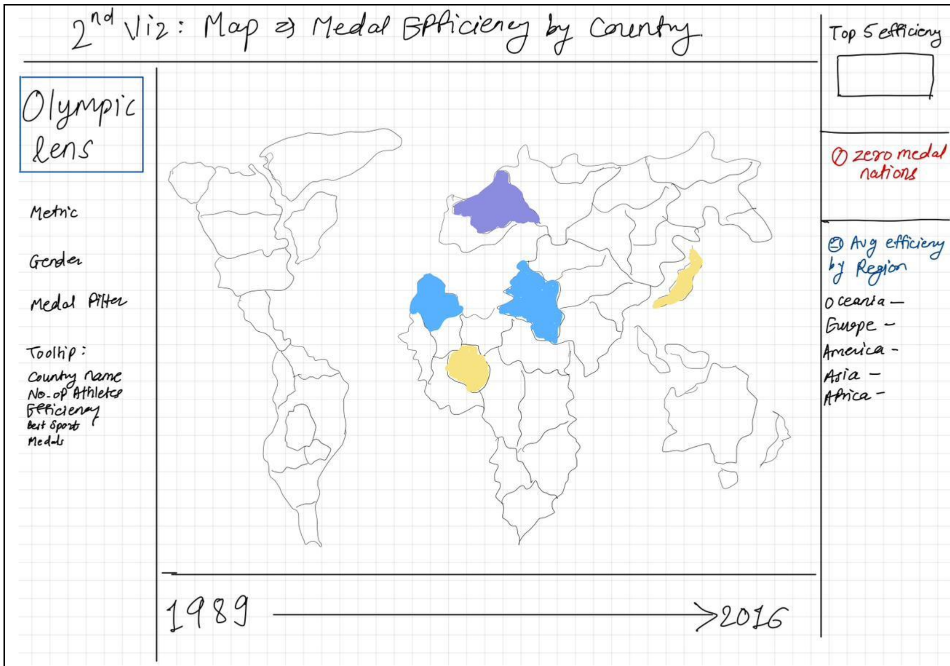


Title: Geopolitics, Wealth, and Glory: What Really Determines Olympic Success Across 120 Years?

Date: 22-06-2026

Research Question: How can interactive visualization of Olympic athlete participation and medal outcomes (1896–2016), contextualized with major world events and gender inclusion data, reveal patterns of geopolitical influence, regional disparity, and structural inequality within the modern Olympic movement?

Idea: Petra Isenberg

	
Focus The most novel aspect of this project is its **layered narrative structure**: three visualizations that each answer their own question but collectively build a single argument. Viz 1 (Timeline) reveals how geopolitical shocks — wars, boycotts, colonial politics — disrupted regional medal distributions. Viz 2 (Choropleth) reframes performance using a **medal efficiency ratio** (medals ÷ athletes sent) instead of raw count, which exposes spatial inequality invisible in standard medal tables and adds a gender participation layer on the same map. Viz 3 (Network) shows how the sports program itself was designed in ways that concentrated dominance — nations already in power benefited most from each new sport added. Together the three visualizations form a **causal chain**: geopolitical disruption → spatial exclusion → sport-level monopoly. Each chart explains the next.	**Detail**

Details:

Datasets:

<u>Primary:</u> 120 Years of Olympic History — Athletes and Results (Kaggle) — 271,116 rows, one row per athlete per event, covering all Summer and Winter Games from Athens 1896 to Rio 2016. Key columns: Name, Sex, Age, Team, NOC, Year, Sport, Event, Medal.

<u>Secondary 1</u>: World Important Events — Ancient to Modern (Kaggle) — major dated events (wars, boycotts, political crises) filtered to 1896–2016, used as overlay markers on visualizations.

<u>Secondary 2:</u> UCDP/PRIO Armed Conflict Dataset v26.1 (Uppsala Conflict Data Program) — 2,303 conflict-year observations covering armed conflicts globally from 1896–2016. Filtered to full-scale wars only (intensity level 2) and deduplicated to 127 unique conflicts by conflict ID. Key columns: conflict_id, location, side_a, side_b, year, intensity_level, type_of_conflict. Used alongside Secondary 1 to enrich geopolitical context in visualizations.

Key metrics defined:

- Medal Efficiency = medals won ÷ athletes sent (per nation per decade) — used in Viz 2
- Women's Participation Rate = female athletes ÷ total athletes sent (per nation per decade) — gender toggle in Viz 2
- Edge weight in network = medal count per country–sport pair per era — used in Viz 3

Colour scheme: Light theme — warm background #f4f2ed, white chart canvas. Continents follow official Olympic ring colours: Europe = #0081C8, Americas = #EE334E, Asia = #FCB131, Africa = #00A651, Oceania = #6F4E9C. Event markers: War red #b91c1c, Boycott orange #c2410c, Host blue #0369a1.

Optional-- Glyphs: Used where multiple data dimensions share one spatial position — dot plot dots encode edition anomalies (Viz 1), podium glyphs show Gold/Silver/Bronze composition on top efficiency nations (Viz 2), split-ring arc glyphs on network nodes show sport-category breadth per country (Viz 3).

Thin data treatment: Nations or regions with fewer than 5 athletes in an edition are shown with a hatched fill overlay — data is never hidden but visually flagged as statistically sparse.

Interaction: Brush-to-zoom on timeline, animated decade stepping with play button on map, era switching with node fade-in/out on network, cross-links between all three visualizations.

Algorithms needed: Group-by aggregation and percentage normalization (Viz 1), efficiency ratio calculation and decade binning with ISO country code join (Viz 2), bipartite graph construction and force-directed layout with edge threshold filtering (Viz 3).

Story:

The Olympic Games are the world's most watched sporting event — yet the standard medal table tells almost nothing about why nations win or lose. It rewards size and wealth and completely obscures the structural forces that determined who could even compete in the first place.

This project starts from one observation: the medal table is a misleading summary of Olympic performance. A nation that sent 500 athletes and won 10 medals looks identical to one that sent 15 athletes and won 10 — even though the second achieved something far more extraordinary.

Using 120 years of athlete-level Olympic data (1896–2016) alongside a world events timeline, this project investigates three interconnected patterns:

Geopolitical shocks did not affect all regions equally. Wars and boycotts disrupted the Games nine times. Europe and North America recovered fastest — not because their athletes were better, but because they already had funded infrastructure and stable institutions. Africa and Asia, still under colonial rule for much of the early 20th century, did not.

Standard medal counts hide inequality. Reframing performance as medal efficiency (medals won ÷ athletes sent) makes invisible nations visible. Jamaica achieves roughly eight times the global average efficiency. The USA's raw dominance reflects the size of its delegation — a luxury only wealthy states can afford.

The Olympic sport programme was never neutral. Sports were added at moments that benefited the nations with the most IOC power. Basketball's 1936 inclusion favored the USA for decades. Judo was added in Tokyo 1964. Taekwondo became a full medal sport in Seoul 2000. The programme is a historical record of institutional power, not athletic universality.

These three patterns — temporal disruption, spatial inequality, and sport-level monopoly — require three different visualization types to be seen clearly. That is why this project uses an annotated timeline, an efficiency choropleth, and a country–sport network: because the story cannot be told any other way.